

Coding & Solution

Date	23 June 2026
Team ID	SWTID-2026-1377
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/sindhuradevi7-ux/Voice-Based-Concept-Understanding-Analyser
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask (Backend), HTML/CSS/JavaScript (Frontend), Scikit-learn, NLTK, Web Speech API

Key Features Implemented	• Voice-based concept explanation using Web Speech API. • Speech-to-text conversion. • NLP preprocessing and semantic similarity analysis. • TF-IDF vectorization and cosine similarity-based concept evaluation. • Detection of missing concepts and misconceptions. • Mastery score calculation and concept coverage analysis. • Personalized feedback generation. • Flask web application with responsive frontend. • Support for predefined and custom concepts. • Downloadable analysis report.
Pending / Incomplete Features	• Cloud deployment (currently runs on the local Flask development server). • User authentication and profile management. • Performance history tracking across multiple sessions. • Multi-language speech recognition and evaluation support. • Advanced Generative AI model integration for deeper semantic analysis.
Setup / Run Instructions	1. Clone the repository and navigate to the project directory. 2. Create a virtual environment. 3. Install dependencies using pip install -r requirements.txt. 4. Run the application using python app.py. 5. Open the browser and access the Flask application through the local server URL (e.g., http://127.0.0.1:5000/).



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial

5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The **Voice-Based Concept Understanding Analyser (ConceptCheckmate)** successfully integrates **Flask**, **Natural Language Processing (NLP)**, and the **Web Speech API** to evaluate students' conceptual understanding through spoken or typed explanations. The application performs semantic similarity analysis using **TF-IDF vectorization** and **cosine similarity**, identifies missing concepts and misconceptions, calculates a mastery score, and generates personalized feedback. The current implementation is fully functional for local deployment and can be enhanced in future versions by incorporating cloud deployment, user authentication, multilingual support, learning progress tracking, and advanced Generative AI models for more comprehensive concept evaluation.